INDIAN INSTITUTE OF INFORMATION TECHNOLOGY
SONEPAT

भारतीय सूचना प्रौद्योगिकीसंस्थानसोनीपत

Email: sonepatiiit@gmail.com, website: www.iiitsonepat.ac.in

Mid Term Examination 2026

Subject: Communication Skills (HUL204) **Roll no ....**

Semester-II **Time- 2 Hrs.**
M.M.-30 marks

Branch CSE/IT/DSA

Note: All questions are compulsory

1.	Write a note on how barrier to communication is a factor in diminishing effective communication. By using appropriate examples, discuss 'linguistic barrier to communication'.	**(6 marks)**
2.	With reference to communication in workplaces, describe the flow of communication? Evaluate the various types of communication based on the flow of communication.	**(6 marks)**
3.	Elaborating the differences between verbal and non-verbal communication. In your opinion, how significant is non-verbal communication in a professional context.	**(6 marks)**
4.	What is semiotics in communication? With the help of suitable examples, discuss various types of signs in communication.	**(6 marks)**
5.	Write a brief note on communication as a visual. Also discuss Gestalt theory of visual communication.	**(6 marks)**

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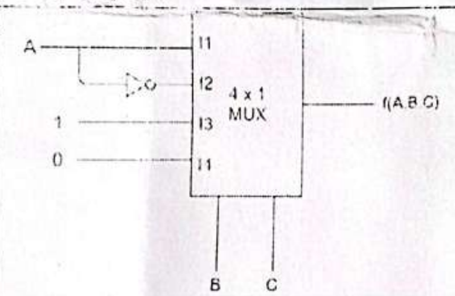
Mid-Semester Examination 2025-26

Semester: II | **Subject: Digital Electronics (ECL-202)** | **Roll no** 12511097...

Branch: CSE/IT/DSA | **M.M.-30 marks** | **Time- 120 Minutes**

Note: All questions are compulsory.

Q-1	Convert the decimal number 473.6 to base-5 number system.	**(3 marks)**
Q-2	Prove the below function using Boolean identities $(wxz + w'x + x'z + y')(y + wx + xz) = x(w + y)(w' + y') + z(x + y)$	**(3 marks)**
Q-3	Subtract the following numbers using 2's complement and 9's complement $(11011)_2 - (01011)_2$	**(3 marks)**
Q-4	A combinational circuit is defined by the following two Boolean functions: $F1(x,y,z) = x'y'z' + xz$ and $F2 = \sum(0,2,4,7)$ Design the combinational circuit that implements the functions with a decoder and external gates.	**(3 marks)**
Q-5	 Analyze the given 4×1 multiplexer circuit and derive the Boolean function represented by it.	**([illegible] marks)**
Q-6	Design a combinational circuit to operate a lamp by using four switches. The lamp lights up if switches 1 and 4 are closed and switch 2 is open, or if 2, 4 are closed and 3 is open, or if all the switches are kept closed. Express this as a Boolean function in minimized SOP and POS form. **Note:** Use bit '1' when switch is closed and bit '0' when switch is open.	**(5 marks)**
Q-7	Minimize the following function using K-Map and draw a circuit diagram using NAND gates for the minimized expression. $F = \sum m(0,2,4,9,12,15) + \sum d(1,5,7,10)$	**(5 marks)**
Q-8	Illustrate the organization of Random Access Memory (RAM) with a diagram and demonstrate how Static RAM (SRAM) and Dynamic RAM (DRAM) differ in structure and performance.	**(5 marks)**

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Mid Semester **Subject: Object Oriented Programming using C++ (UCS004)** **Roll no ....**

Branch: CSE/IT/DS **M.M.-30 marks** **Time- 2 Hours.**

•Attempt all questions.

•Support answers with diagrams, algorithms, and C++ code wherever appropriate.

•Assume suitable data wherever necessary.

Q-1	Object-Oriented Programming aims to improve software design and maintainability. a) Explain Encapsulation, Abstraction, Inheritance, and Polymorphism with suitable examples. b) Analyze how these concepts improve software modularity, maintainability, and security compared to structured programming. c) Discuss two real-world systems where OOP design significantly improves scalability.	**(5 mark**
Q-2	Memory organization is crucial for writing efficient programs. a) Explain how one-dimensional and two-dimensional arrays are stored in memory. b) Derive the address calculation formula for elements of a two-dimensional array in row-major order. c) Given the base address of a 2D array as 1000, element size 4 bytes, and dimensions 4×5, calculate the address of element A[2][3].	**(5 marks**
Q-3	Pointers provide powerful mechanisms for memory access in C++. a) Explain the relationship between arrays and pointers with memory diagrams. b) Demonstrate pointer arithmetic by writing a C++ program that accesses array elements using pointer notation. c) Compare arrays of pointers and pointers to pointers, explaining their memory representation and practical applications.	**(5 marks)**
Q-4	Classes form the foundation of object-oriented programming. a) Explain the concept of classes and objects in C++. b) Discuss how access specifiers (private, public, protected) enforce data hiding and controlled access. c) Design a C++ class 'Employee' that includes employee ID, name, and salary with functions to input and display details and an inline function to compute annual salary. Explain the design decisions.	**(5 marks**
Q-5	Constructors and operator overloading are important features in C++. a) Differentiate between default, parameterized, and copy constructors. b) Explain constructor overloading and its role in object initialization. c) Design a C++ class 'Complex' that overloads the + and – operators to perform addition and subtraction of complex numbers.	**(5 mark**
Q-6	Inheritance and polymorphism enable flexible program design. a) Differentiate between single, multiple, multi-level, and hierarchical inheritance with diagrams. b) Explain virtual functions and how they enable runtime polymorphism. c) Design a C++ program for a banking system with a base class 'Account' and derived classes 'SavingsAccount' and 'CurrentAccount'. Implement function overriding and include exception handling for invalid withdrawals.	**(5 mark**

CamScanner

Roll No....12S11097

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B.Tech. 2nd Sem Mid Semester Examination, 9th March, 2026.

Subject: Engineering Mathematics-II **Subject Code: MAL201**

Branch:CSE/IT/DSA **Max.Marks.-30** **Time- 2 Hours**

Note: All questions are compulsory.

Q-1	Evaluate the ~~triple~~ Double integral $\iint_A x^2 dxdy$, where A is the region in the first quadrant bounded by the hyperbola xy=16 and the lines y=x, y=0 and x=8. 448	3 Marks L5, L6
Q-2	Evaluate the following integral by changing the order of integration $\int_0^a \int_{a-\sqrt{a^2-y^2}}^{a+\sqrt{a^2-y^2}} xy\, dx\, dy$.	3 Marks L3, L5
Q-3	Transform the double integral to polar coordinate and hence evaluate it, $\int_{y=0}^{y=a} \int_{x=0}^{x=\sqrt{a^2-y^2}} (a^2 - x^2 - y^2) dx\, dy$.	3 Marks L2, L3
Q-4	Find the volume of the region D enclosed by the surfaces $z = x^2 + 3y^2$ and $z = 8 - x^2 - y^2$	3 Marks L5, L4
Q-5	Solve the differential equation $(xy^2 - x^2)dx + (3x^2y^2 + x^2y - 2x^3 + y^2)dy = 0$.	3 Marks L2, L6
Q-6	Using Picard's Iteration Method Obtain a Solution Up to 5th approximation of the equation $y' = x + 3y$ such that $y(0) = 1$. Also compare the result with the exact solution.	3 Marks L3,L6
Q-7 8	Find the Complete Solution of the Differential Equation: $(4D^2 + 8D + 3)y = xe^{-\frac{x}{2}} cos x$	3 Marks L1, L5
Q-8	Find the current *I(t)* in an RLC-circuit with R=11Ω(ohms),L=0.1H(henry), C=10^{-2} F(farad),which is connected to a source of EMF E(t) =110 sin (60.2πt)=110 sin 377t. Assume that current and capacitor charge are 0 when t=0. (Kirchhoff's Voltage Law $LI'' + RI' + \frac{1}{C}I = E(t)$)	6Marks L2, L5
Q-9	Solve the following system of linear DE equations: $Dx + y = \sin t$ & $Dy + x = \cos t$, Given that $x(0) = 2$ & $y(0) = 0$.	3Marks L1, L2

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Mid Sem **Subject Name: Web Designing** **Subject code: UIT205/UCS003** **Roll no** 12511039
Branch: CSE/IT/DSA **Semester : 2** **M.M.-30 marks** **Time- 2 Hours**

Note: All questions are compulsory.

Q-1	A website's homepage has a complex interactive animation that slows down page load time. The client loves the animation but wants the site to load fast. How do you resolve this? What are the basic principles involved in developing a website?	**(6 marks)**
Q-2	Web standards play a crucial role in ensuring consistency, accessibility, and compatibility across different web platforms and devices. In this context, explain the importance of web standards in modern web development. Discuss the role of the World Wide Web Consortium (W3C) in establishing and maintaining these standards. Additionally, explain the concept of responsive web design and describe how techniques such as flexible layouts, media queries, and responsive images help in developing websites that adapt effectively to different screen sizes and devices.	**(6 marks)**
Q-3	Differentiate between static and dynamic websites. Explain their architecture, technologies used, advantages, limitations, and suitable use cases.	**(6 marks)**
Q-4	(a) HTML provides several elements for organizing and presenting content effectively. Discuss the different text formatting elements in HTML, including headings, paragraphs, line breaks, and text styling tags. Also explain the different types of lists in HTML such as ordered lists, unordered lists, and definition lists with examples. (b) Create an HTML table with a border, showing a product inventory with columns for Product ID, Name, Price, and Quantity.	**(6 marks)**
Q-5	(a) Explain the structure and working of HTML forms and describe common form controls such as text fields, radio buttons, checkboxes, dropdown menus, and submit buttons. Also briefly discuss the concept of frames in HTML, including their purpose, advantages, and limitations in modern web development. (b) Design an HTML form for a feedback page with fields for Name, Email, Feedback Type (dropdown with options like Complaint, Suggestion, Query), and a Message box. Include a Submit button.	**(6 marks)**

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Mid Term Examination - March 2026

Subject Name: Data Structure **Subject Code: UCS002** **Roll no** 12511039

Branch: CSE/IT/DSA **Semester 2nd** **M.M.- 30 marks** **Time- 120 min.**

Note: Attempt all questions in sequence.

Q-1	A	A function fun takes two arguments: an integer k and a queue of integers. The function should reverse the order of the first k elements of the queue, while keeping the remaining elements in the same relative order. The partial code for the function is given below. Fill in the missing lines of code (1–5) to correctly implement the required functionality. What is the **base condition** used in the solve() function and why is it necessary?
		`fun (queue q, int k) {`
		`  solve(q, k);`
		`  int s = q.size() – k;`
		`  while (s–> 0) {`
		`    int x = q.front();`
		`    ______1______;`
		`    ______2______;`
		`  }`
		`  return q;`
		`}`
		`void solve(queue &q, int k) {`
		`  if (k == 0) return;`
		`  ______3______;`
		`  ______4______;`
		`  solve(q, k - 1);`
		`  ______5______;`
		`}`
			(6)

	B	Suppose an initially empty stack S has executed a total of n push operations, n/3 top operations, n/6 peek operation and n/2 pop operations, n/6 of which raised empty errors that were caught and ignored. What is the current size of S?	(4)
	C	Given the base address of a two-dimensional array $A[21..48][100..150]$ as 2000, and the size of each element is 2 bytes in memory. The array is stored in row-major order. Find the memory address of the element $A[43][120]$.	
Q-2		Consider the following list of integers: 7, 12, 9, 11, 3, 21, 5	10
	A	Write the complete pseudocode for the Quick Sort algorithm using the Divide and Conquer strategy.	
	B	Using Quick Sort with the first element as the pivot, perform the sorting of the above list. Show each partitioning step clearly, including the position of the pivot after every partition.	
	C	Analyze the time complexity of Quick Sort for the following cases and justify your answer: Best Case, Average Case, Worst Case	
	D	Explain two different pivot selection strategies and discuss how they affect the performance of Quick Sort.	
	E	Is Quick Sort a stable sorting algorithm? Justify your answer	
Q-3		Write a C program to create a singly linked list. Define and implement a function that moves the last node of the linked list to the beginning of the list. The function should modify the original list and return the updated list. Also, mention real-life applications of this operation.	10

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